

LAB REPORT

Lab Report No: 01



FCFS Implementation

Course Information

Course Title: Operating System Lab
Course Code: CSE 402
Department: Computer Science and Engineering
University: University of Asia Pacific

Submitted By

Name: Faiaz Masnun Labib
Student ID: 23101128
Section: C-1

Submitted To

Instructor Name: Atia Rahman
Orthi
Designation: Lecturer
Department: CSE Department

Date of Submission: August 13, 2026

1. Problem Statement:

Implement CPU Scheduling for the following scenario where the number of processes = 5 and the process information is given below:

Process ID	Arrival Time	Burst Time
P1	3	5
P2	2	4
P3	4	3
P4	1	2
P5	5	3

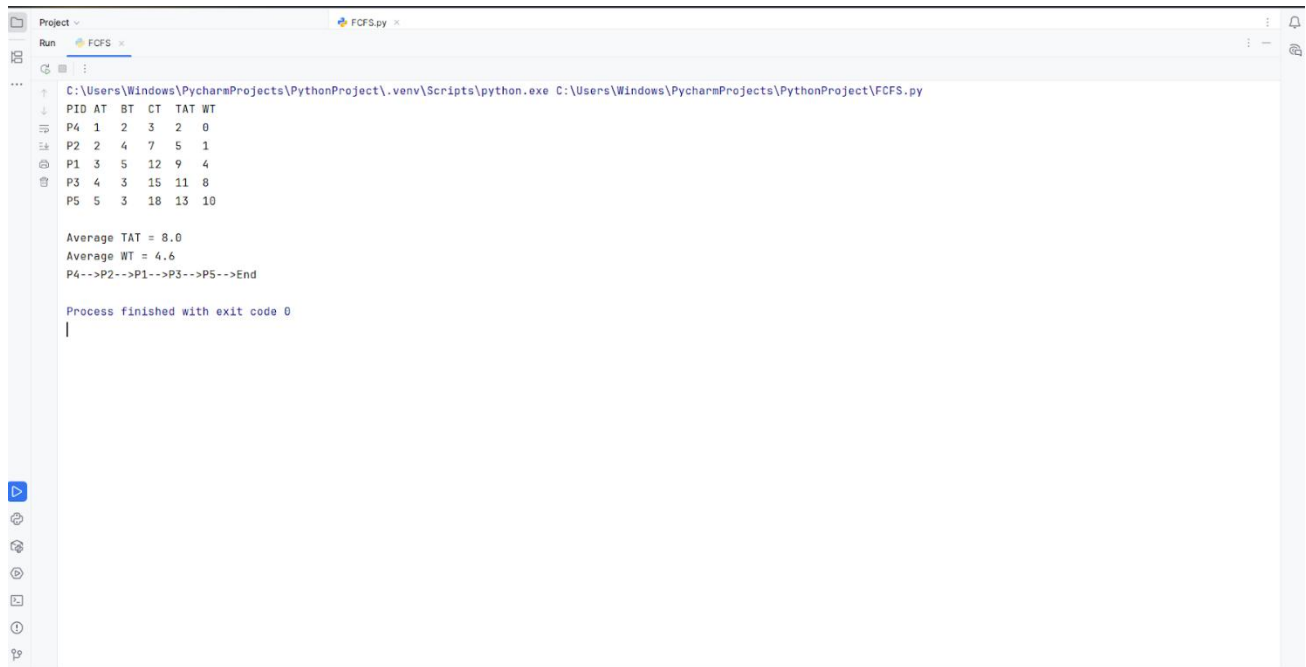
Calculate Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), Average TAT, and Average WT. Also represent the process execution sequence.

2. Source Code:

```
1 processes = [  
2     ["P1", 3, 5],  
3     ["P2", 2, 4],  
4     ["P3", 4, 3],  
5     ["P4", 1, 2],  
6     ["P5", 1, 3]  
7 ]  
8  
9 processes.sort(key=lambda x: x[1])  
10  
11 time = 0  
12 total_tat = 0  
13 total_wt = 0  
14  
15 print("PID\tAT\tBT\tCT\tTAT\tWT")  
16  
17 for p in processes:  
18     pid, at, bt = p  
19  
20     if time < at:  
21         time = at  
22  
23     ct = time + bt  
24     tat = ct - at  
25     wt = tat - bt  
26  
27     total_tat += tat  
28     total_wt += wt  
29  
30     print(f"{pid}\t{at}\t{bt}\t{ct}\t{tat}\t{wt}")  
31     time = ct  
32  
33 n = len(processes)  
34  
35 print(f"\nAverage TAT = {total_tat / n:.2f}")  
36 print(f"Average WT = {total_wt / n:.2f}")  
37  
38 print("\nExecution Sequence:")  
39 for p in processes:  
40     print(p[0], end=" -> ")  
41 print("End")
```

GitHub: [Click Here!!!!!!](#)

3. Result:



```
C:\Users\Windows\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Windows\PycharmProjects\PythonProject\FCFS.py
PID AT  BT  CT  TAT WT
P4 1  2  3  2  0
P2 2  4  7  5  1
P1 3  5  12 9  4
P3 4  3  15 11 8
P5 5  3  18 13 10

Average TAT = 8.0
Average WT = 4.6
P4-->P2-->P1-->P3-->P5-->End

Process finished with exit code 0
|
```

4. Discussion:

- **Algorithm:** Implemented non-preemptive First-Come, First-Served (FCFS) CPU scheduling for 5 processes.
- **Execution Order:** Sorted by arrival time, resulting in the sequence **P4 → P2 → P1 → P3 → P5**.
- **CPU Idle Time:** CPU was idle from time 0 to 1 as the first process (P4) arrived at time 1.
- **Core Formulas:**
 - Turnaround Time: $TAT = CT - AT$
 - Waiting Time: $WT = TAT - BT$
- **Outputs:**
 - **Average Turnaround Time:** 8.0 time units
 - **Average Waiting Time:** 4.6 time units